

Probability fundamentals

COMPLETE LEARNING GUIDE

What You'll Learn

Probability is the mathematical language for reasoning about uncertainty, and it underlies almost everything downstream in this course — from how a classifier's predicted 'confidence' should be interpreted to how a hypothesis test decides whether an effect is real. This submodule covers the basic rules of probability (how probabilities combine for independent and mutually exclusive events), conditional probability (how new information changes the probability of an event), and Bayes' theorem — arguably the single most important formula in this entire submodule, since it's the formal tool for updating a belief when new evidence arrives. Every rule is paired with a fully worked numeric example, including the classic medical-testing example that shows why a 'highly accurate' test can still produce mostly false positives when the condition it's testing for is rare.

1. Sample Spaces, Events, and the Basic Rules

A sample space is the set of all possible outcomes of a random process (e.g. for one die roll, $\{1,2,3,4,5,6\}$); an event is any subset of that space (e.g. 'rolling an even number' = $\{2,4,6\}$). Every probability is a number between 0 (impossible) and 1 (certain), and the probabilities of all outcomes in a sample space always sum to 1. The addition rule for mutually exclusive events (events that can't both happen) is $P(A \text{ or } B) = P(A) + P(B)$; for events that can overlap, you must subtract the overlap: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$, to avoid double-counting outcomes that satisfy both.

- Rolling a die: $P(2) = 1/6$, and $P(\text{even}) = P(2)+P(4)+P(6) = 3/6 = 0.5$, since these three outcomes are mutually exclusive.
- Drawing a card: $P(\text{King or Heart}) = P(\text{King}) + P(\text{Heart}) - P(\text{King of Hearts}) = 4/52 + 13/52 - 1/52 = 16/52 \approx 0.308$ — the overlap (King of Hearts) must be subtracted once.
- For independent events, $P(A \text{ and } B) = P(A) \times P(B)$ — e.g. two fair coin flips both landing heads: $0.5 \times 0.5 = 0.25$.

2. Independence vs. Dependence

Two events are independent if the occurrence of one has no effect on the probability of the other (successive fair coin flips are independent — the coin has no memory). Events are dependent when one outcome changes the probability of the other — drawing cards without replacement is the classic example, because removing one card changes the composition of the remaining deck. Correctly identifying independence is essential before applying the simple multiplication rule $P(A \text{ and } B) = P(A) \times P(B)$, which only holds for independent events; for dependent events you need conditional probability instead.

- Coin flips, dice rolls, and (with replacement) card draws are independent — past outcomes don't change future probabilities.
- Drawing two cards without replacement is dependent: $P(\text{2nd card is a King} \mid \text{1st card was a King}) = 3/51$, not $4/52$, because one King is already gone.
- Multiplying probabilities as if events were independent when they're actually dependent is one of the most common probability errors in practice.

3. Conditional Probability: Updating on New Information

Conditional probability, written $P(A \mid B)$ and read 'the probability of A given B,' is the probability of event A occurring given that event B is already known to have occurred. It's formally defined as $P(A \mid B) = P(A \text{ and } B) / P(B)$ — restricting the sample space down to only the outcomes where B happened, then asking what fraction of those also satisfy A. Conditional probability is the mathematical backbone of Bayes' theorem, and intuitively it's just 'given what I now know, how likely is this?' — the same reasoning you use informally whenever you update an expectation based on new evidence.

- Worked example: in a deck of 52 cards, $P(\text{King and Heart}) = 1/52$. $P(\text{Heart}) = 13/52$. So $P(\text{King} \mid \text{Heart}) = (1/52) / (13/52) = 1/13$ — consistent with there being exactly one King among the 13 hearts.
- If A and B are independent, $P(A \mid B) = P(A)$ — knowing B happened tells you nothing new about A, by definition of independence.

4. Bayes' Theorem: The Formula for Updating Beliefs

Bayes' theorem rearranges the definition of conditional probability to let you compute $P(A \mid B)$ when you actually know $P(B \mid A)$ instead — extremely common in practice, because it's often far easier to measure 'given the disease, how often does the test come back positive' (a property of the test) than to directly measure 'given a positive test, how likely is the disease' (what you actually want to know). The formula is: $P(A \mid B) = [P(B \mid A) \times P(A)] / P(B)$, where $P(A)$ is the prior probability (belief before seeing evidence B), $P(B \mid A)$ is the likelihood, and $P(A \mid B)$ is the posterior (updated belief after seeing the evidence).

PYTHON

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# Bayes' theorem for the classic medical testing scenario
prior_disease = 0.01          # 1% of the population has the disease
sensitivity = 0.99            # P(positive test | disease) - true positive rate
false_positive_rate = 0.05    # P(positive test | no disease)

p_positive = (sensitivity * prior_disease) + (false_positive_rate * (1 -
prior_disease))
p_disease_given_positive = (sensitivity * prior_disease) / p_positive

print(round(p_disease_given_positive, 4)) # 0.1667
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5. Worked Example: Why a 99% Accurate Test Can Still Mislead

Continuing the code example above by hand: assume a disease affects 1% of the population (prior = 0.01), a test correctly identifies 99% of people who actually have it (sensitivity = 0.99), and incorrectly flags 5% of healthy people as positive (false positive rate = 0.05). First compute the overall probability of testing positive, accounting for both true and false positives: $P(\text{positive}) = (0.99 \times 0.01) + (0.05 \times 0.99) = 0.0099 + 0.0495 = 0.0594$. Then apply Bayes' theorem: $P(\text{disease} \mid \text{positive}) = 0.0099 / 0.0594 \approx 0.167$, or about 16.7%. Despite the test being '99% accurate' on people who have the disease, a random person who tests positive only actually has the disease about 1 time in 6 — because the disease is rare, false positives from the large healthy population outnumber true positives from the small sick population.

- This result is famously counter-intuitive and is exactly why screening-test results are always interpreted alongside the base rate (prior), never in isolation.
- The lower the prior (rarer the condition), the more a fixed false-positive rate dominates the result — this is why rare-disease screening tests are typically followed by a confirmatory second test.

6. Where This Shows Up Later in the Course

Bayes' theorem isn't just a standalone puzzle — it's the conceptual foundation of an entire family of ML models (Naive Bayes classifiers), it's how spam filters historically worked (updating the probability a message is spam based on which words appear), and it's the correct lens for interpreting almost any diagnostic or classification system's output: a model's raw 'positive' prediction is a likelihood-like signal, and the actual probability you care about also depends on how common the thing you're predicting is in the first place.

- A fraud-detection model flagging a transaction as 'fraud' should be interpreted the same way as the medical test example — the rarity of actual fraud matters as much as the model's individual accuracy.
- Naive Bayes classifiers (covered later) apply this exact formula, treating each input feature as new evidence that updates the probability of each class.

Common Pitfalls

- Multiplying $P(A) \times P(B)$ to get $P(A \text{ and } B)$ when the events are actually dependent, not independent.
- Forgetting to subtract the overlap $P(A \text{ and } B)$ when computing $P(A \text{ or } B)$ for events that can occur together.
- Confusing $P(A \mid B)$ with $P(B \mid A)$ — these are generally different numbers, and mixing them up is the single most common probability mistake (it's exactly the error Bayes' theorem exists to correct).
- Ignoring the base rate (prior) when interpreting a positive test or model prediction, leading to overconfidence in a rare-condition scenario.

- Treating a 'highly accurate' test/model as equivalent to 'a positive result is highly likely to be correct' — these are only the same when the condition being detected is common.
- Assuming successive real-world events (e.g. card draws without replacement, or correlated sensor readings) are independent when they're actually dependent.

Key Takeaways

- $P(A \text{ or } B)$ needs an overlap correction unless the events are mutually exclusive; $P(A \text{ and } B) = P(A) \times P(B)$ only holds for independent events.
- Conditional probability $P(A|B) = P(A \text{ and } B)/P(B)$ formalizes 'given what I now know, how likely is this?'
- Bayes' theorem lets you flip a known conditional probability ($P(B|A)$) around to get the one you actually want ($P(A|B)$).
- A rare condition plus an imperfect test almost always means most positive results are false positives — the base rate matters as much as the test's accuracy.
- $P(A|B)$ and $P(B|A)$ are not the same number — confusing them is the single most common real-world probability error.

Learn More

Khan Academy — Probability

<https://www.khanacademy.org/math/statistics-probability/probability-library>

Stanford Encyclopedia of Philosophy — Bayes' Theorem

<https://plato.stanford.edu/entries/bayes-theorem/>

Seeing Theory — A Visual Introduction to Probability

<https://seeing-theory.brown.edu/>